

SDA Assignment # 1

**Cosat University Islamabad
Abbottabad Campus**



INSTRUCTOR: MUKHTIAR ZAMIN SIR

SUBJECT: SOFTWARE DESIGN AND ARCHITECTURE

SUBMITTED BY: MAAZ ALI KHAN

REG NO: SP23-BSE-052

Section: BSE-5B

Case Study: Student Feedback Module

Overview

The Student Feedback Module is a web-based system designed to collect anonymous feedback from students about faculty and campus services. It allows secure login, feedback submission, and admin report access.

Purpose

To improve teaching quality and campus services through structured student feedback.

Core Features

- Student registration/login (JWT-based)
- Faculty search and selection
- Rating-based feedback form (10 questions + comments)
- Admin report viewing and question management

Artifacts Created

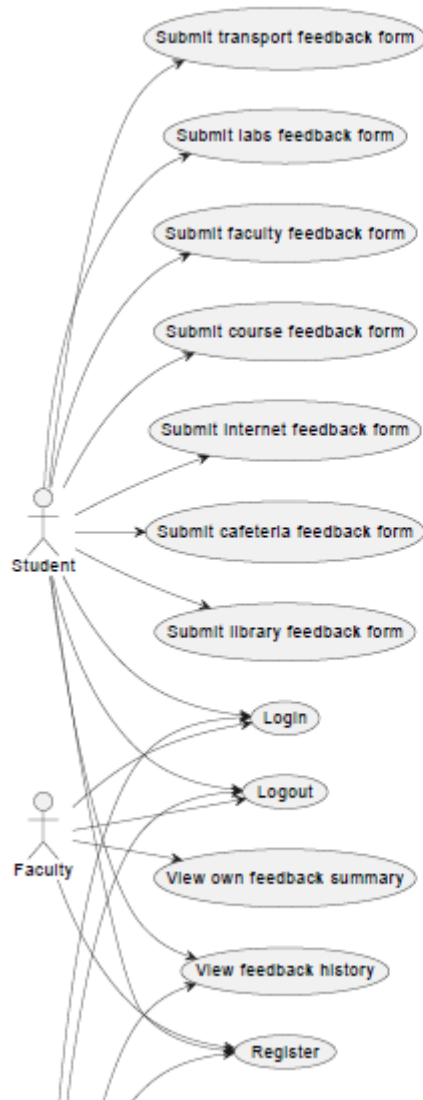
- **Class Diagram:** Shows relationships among Student, Faculty, Feedback, Admin, and Services.
- **Use Case Diagram:** Actors include Student and Admin; main use cases are Login, Submit Feedback, and View Reports.
- **Communication Diagram:** Describes how Student UI, Controller, AuthMiddleware, and DB interact during feedback submission.
- **System Sequence Diagram (SSD):** Illustrates the sequence of actions in submitting feedback.
- **Fully Dressed Use Case:** Submit Feedback — includes goal, actor, steps, and extensions.
- **Package Diagram:** Organized code into auth, feedback, models, routes, and frontend components.

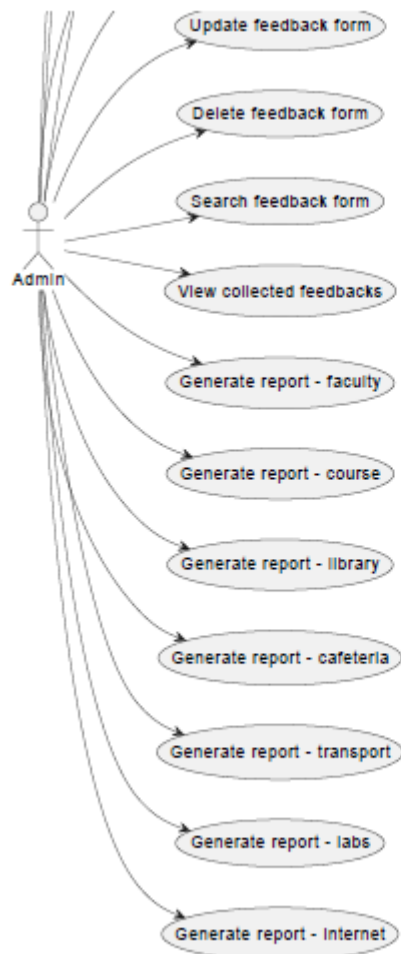
Conclusion

The module ensures anonymous, reliable feedback collection with structured components and modular design for easy maintenance and future scalability.

Let me know if you want it in a presentation format or need any visuals.

USE CASE DAIGRAM:





FULLY DRESSED USE CASE:

Use Case: View Faculty Feedback Form

Use Case Name	submit Faculty Feedback Form
Actor	Student
Description	This use case allows the student to search faculty member and fill the form and submit the feedback.
Trigger	The student selects the option to submit faculty feedback from the system dashboard.
Preconditions	- Actor is registered and logged in. - Feedback data exists in the database.
Postconditions	- Feedback data is successfully displayed. - Actor may view average ratings and comments.

	-Feedback data is successfully submitted and message display Thank you for your feedback.
--	--

Main Success Scenario (Basic Flow):

1. 1.Student is registered and logged in.
2. 2. Actor clicks “Submit Faculty Feedback” option at the dashboard.
3. 3. System displays a searchable list or dropdown of faculty members.
4. 4. Actor selects a faculty member.
5. 5. System retrieves related feedback from the database.
6. 6. System displays:
 - All 10 question ratings (1–5 or stars)
 - Optional student comments
 - Average rating per question (optional)
7. 7. Actor logs out or navigates away.

Extensions (Alternate Flows):

- No feedback available: System displays a message: “No feedback available for this faculty.”
- Database error: System displays an error message: “Unable to retrieve feedback at this time.”

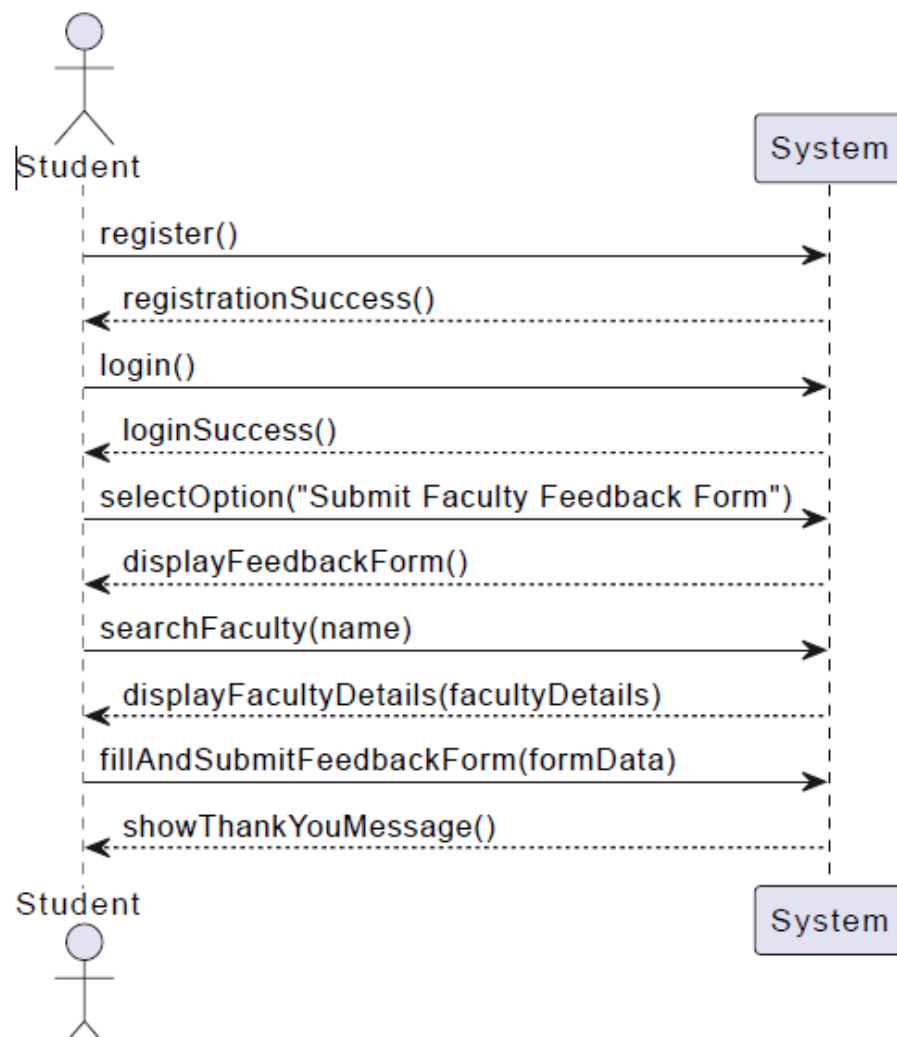
Frequency of Use:

Weekly or after semester-end evaluations.

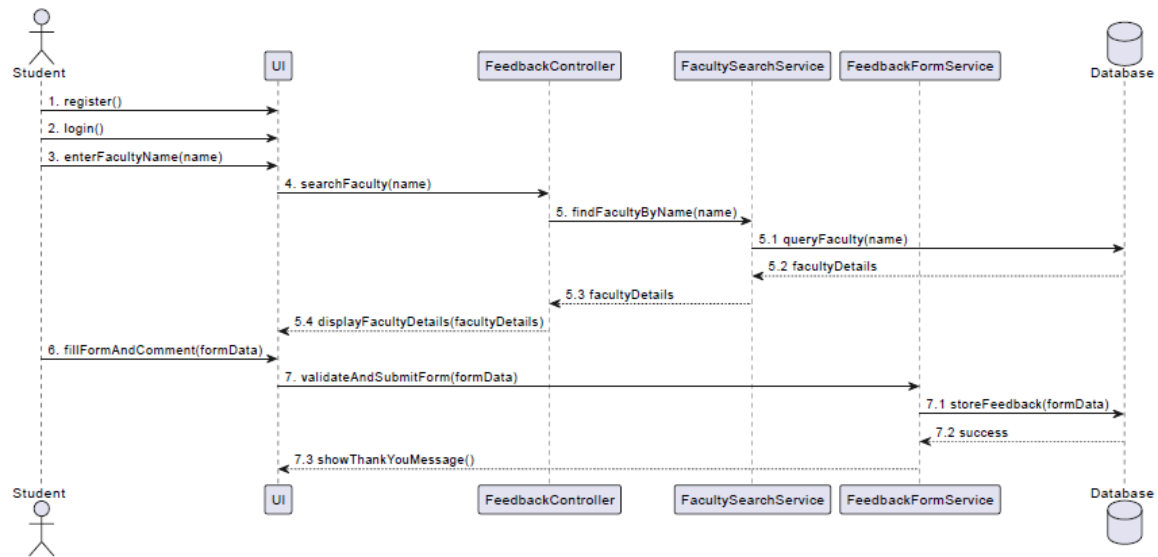
Assumptions:

- Faculty list is preloaded and maintained.
- Role-based access control ensures only authorized users can access feedback.

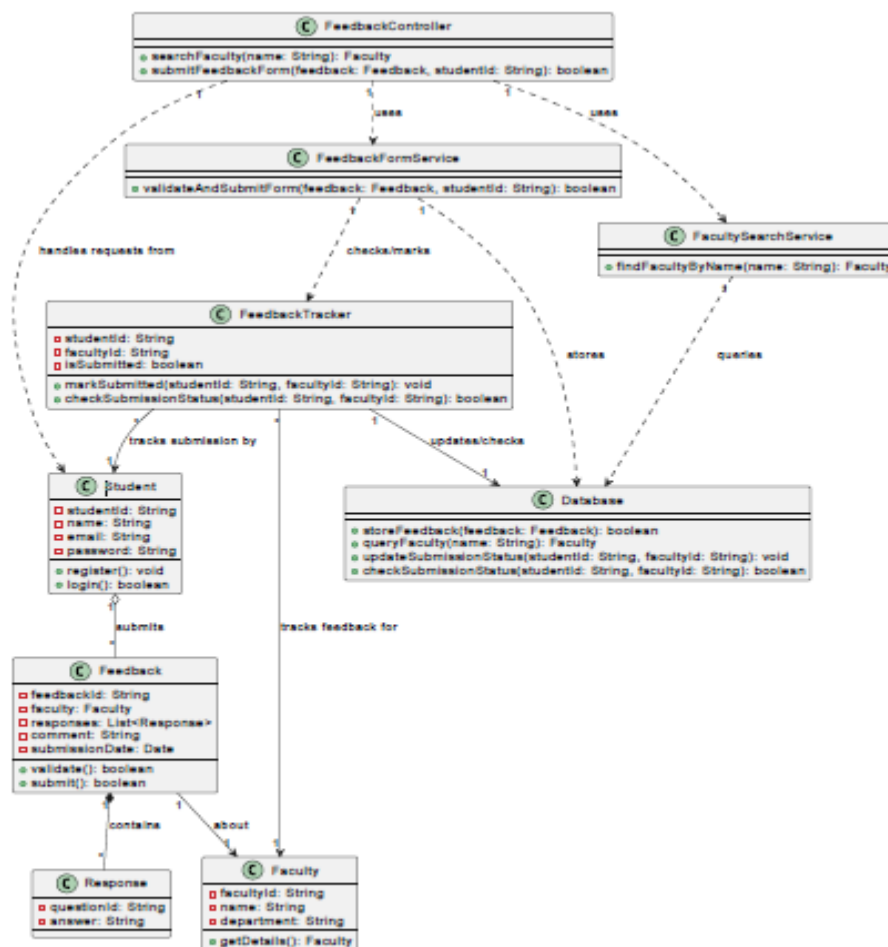
SSD:



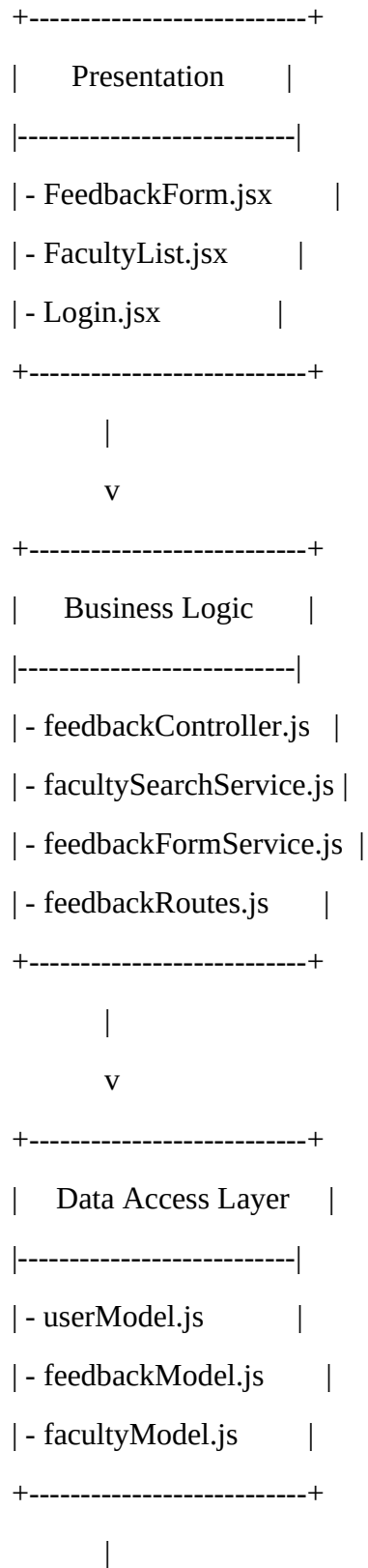
COMMUNICATION DAIGRAM:



CLASS DAIGRAM:



PACKAGE DAIGRAM:



v

+-----+

| Database |

|-----|

| - MongoDB |

+-----+